

House Price Prediction using Feature Engineering & Model Optimization

Artificial Intelligence & Machine Learning – Task 2

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1. Objective

The objective of this project is to improve the performance of a house price prediction model by applying feature engineering techniques and comparing multiple regression algorithms. Feature scaling is performed using StandardScaler, and three regression models are trained and evaluated to identify the best-performing model.

2. Introduction

Machine Learning is widely used to solve regression problems where the objective is to predict continuous values. In this project, house prices are predicted using the California Housing Dataset.

Unlike Task 1, this project focuses on improving model performance through feature engineering and comparing multiple regression algorithms.

The following regression models were implemented:

- Linear Regression
- Ridge Regression
- Decision Tree Regressor

Their performance was compared using MAE, RMSE, and R^2 Score.

3. Dataset Description

The project uses the **California Housing Dataset** available in the Scikit-learn library.

The dataset contains housing-related information collected from California districts.

Features Used

Feature	Description
MedInc	Median Income
HouseAge	House Age
AveRooms	Average Rooms
AveBedrms	Average Bedrooms
Population	Population
AveOccup	Average Occupancy
Latitude	Latitude
Longitude	Longitude
HousePrice	Target Variable

Insert Screenshot: Dataset Preview

```
# Load the dataset
housing = fetch_california_housing()

# Convert it into a DataFrame
df = pd.DataFrame(
    housing.data,
    columns=housing.feature_names
)

# Add the target column
df["HousePrice"] = housing.target

print("✅ Dataset loaded successfully!")

# Display the first 5 rows
df.head()
```

4. Feature Engineering

Feature Engineering improves the quality of data before training machine learning models.

In this project, **StandardScaler** was used to standardize all numerical features.

StandardScaler transforms the dataset so that:

- Mean = 0
- Standard Deviation = 1

Feature scaling helps machine learning algorithms perform more effectively when features have different ranges.

5. Data Preprocessing

The following preprocessing steps were performed:

- Imported required libraries
 - Loaded California Housing Dataset
 - Converted dataset into a Pandas DataFrame
 - Performed Exploratory Data Analysis (EDA)
 - Checked missing values
 - Applied Feature Scaling using StandardScaler
 - Split the dataset into:
 - 80% Training Data
 - 20% Testing Data
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6. Model Development

Three machine learning regression algorithms were trained.

Linear Regression

Linear Regression establishes a linear relationship between the independent variables and the target variable.

Advantages

- Simple
 - Fast
 - Easy to interpret
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Ridge Regression

Ridge Regression is an extension of Linear Regression that applies L2 Regularization to reduce overfitting.

Advantages

- Better generalization
 - Handles multicollinearity
 - Reduces overfitting
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Decision Tree Regressor

Decision Tree Regression predicts values by learning decision rules from training data.

Advantages

- Handles non-linear relationships
 - Easy to visualize
 - No feature scaling required (though scaling was applied consistently in this project)
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7. Model Performance Comparison

The trained models were evaluated using:

- Mean Absolute Error (MAE)
- Root Mean Squared Error (RMSE)
- R^2 Score

Insert Screenshot:

- Model Comparison Table

Model Comparison

	Model	MAE	RMSE	R2 Score
2	Decision Tree	0.433097	0.644279	0.683233
1	Ridge Regression	0.533193	0.745554	0.575819
0	Linear Regression	0.533200	0.745581	0.575788

8. Results and Visualization

The performance of all three models was compared.

The best-performing model was selected based on:

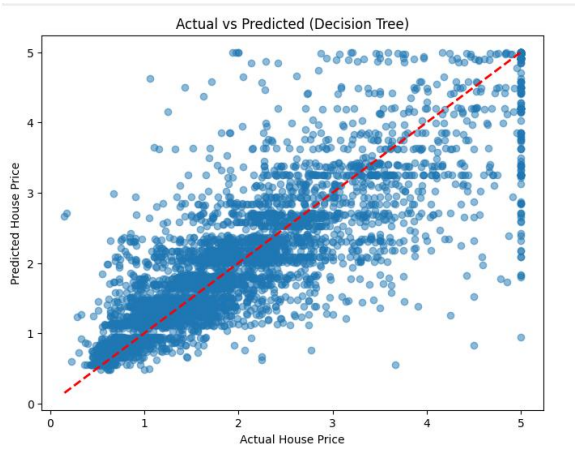
- Highest R^2 Score
- Lowest RMSE
- Lowest MAE

An **Actual vs Predicted** scatter plot was generated to visualize the prediction accuracy.

The closer the points are to the reference line, the better the prediction performance.

Insert Screenshot:

- Actual vs Predicted Graph



9. Future Improvements

The performance of the project can be further improved by using:

- Random Forest Regression
 - Gradient Boosting Regression
 - XGBoost
 - Hyperparameter Tuning
 - Cross Validation
 - Feature Selection
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10. Conclusion

This project successfully demonstrated Feature Engineering and Model Optimization techniques for house price prediction.

The California Housing Dataset was preprocessed using StandardScaler, and three regression algorithms—Linear Regression, Ridge Regression, and Decision Tree Regressor—were trained and evaluated.

The models were compared using MAE, RMSE, and R^2 Score, and the best-performing model was selected based on its evaluation results.

This project highlights the importance of preprocessing, model comparison, and performance evaluation in building effective machine learning solutions.

11. References

1. Scikit-learn Documentation
2. California Housing Dataset
3. Python Documentation
4. Pandas Documentation
5. NumPy Documentation
6. Matplotlib Documentation